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Roll No

BE-3001 (EE/EX)-CBGS

B.E., III Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Mathematics - III

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Express the function $f(x) = x \sin x$, as a Fourier series in $-\pi \leq x \leq \pi$. Hence deduce that

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots = \frac{\pi - 2}{4} \dots \quad 7$$

- b) Find the Fourier sine transform of $f(x) = e^{-ax}/x$. 5

- c) State and prove Linearity property of Fourier transform. 2

2. a) Solve the integral equation

$$\int_0^{\infty} f(x) \cos sx \, dx = \begin{cases} 1-s, & 0 \leq s \leq 1 \\ 0, & s > 1 \end{cases}$$

Hence prove that $\int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$. 7

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- b) Find the Laplace transform of $\frac{1 - \cos t}{t^2}$ 7
3. a) Find the inverse Laplace transform of $\frac{1}{S^4 + 4}$ 7
- b) If $u - v = (x - y)(x^2 + 4xy + y^2)$ and $f(z) = u + iv$ is an analytic function of z . Find $f(z)$. 7
4. a) Evaluate $\int_C \frac{z^2 + 1}{z^2 - 1} dz$ where C is 7
- i) $|z| = 3/2$
- ii) $|z| = 1/2$
- iii) $|z - 1| = 1$
- b) If the directional derivative of $\phi = ax^2y + by^2z + cz^2x$ at the point $(1, 1, 1)$ has maximum magnitude 15 in the direction of $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$, find a, b and c. 7
5. a) Evaluate surface integral $\iint_S \vec{F} \cdot \hat{n} \, ds$, where $\vec{F} = (x^2 + y^2 + z^2)(\hat{i} + \hat{j} + \hat{k})$, S is the surface of the tetrahedron $x = 0, y = 0, z = 0, x + y + z = 2$ and $\hat{n}$ is the unit normal in the outward direction to the closed surface S. 7

[3]

- b) Find the Fourier series for $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$

Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$ 7

6. a) A periodic function of period 4 is defined as $f(x) = |x|, -2 < x < 2$. Find its Fourier series expansion. 7

- b) Find the Fourier cosine transform of $\frac{1}{1+x^2}$. 7

7. a) Find the Fourier sine transform of $\frac{1}{x}$. 3

- b) State and prove modulation property of Fourier transform. 4

- c) Solve $y'' + 2y' + 5y = e^{-x} \sin x$ where $y(0) = 0, y'(0) = 1$ by Laplace transform method. 7

8. a) Use convolution theorem to find $L^{-1} \left[\frac{1}{(s^2 + a^2)^2} \right]$ 7

- b) Evaluate $\int_{-\infty}^{\infty} \frac{x \sin \pi x}{x^2 + 2x + 5} dx$ by contour integration. 7

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